Trainings

General Information

Cummins Inc. recommends the use of fully-formulated antifreeze or coolant containing a precharge of supplemental coolant additive (SCA). The antifreeze or coolant **must** meet the specifications outlined in the Technology and Maintenance Council (TMC) Recommended Practice (RP) 329 (ethylene glycol) or Recommended Practice (RP) 330 (propylene glycol). The use of fully-formulated antifreeze or coolant significantly simplifies cooling system maintenance.

Copies of Technology and Maintenance Council (TMC) specifications can be obtained through Cummins Inc., or by contacting:

Technology and Maintenance Council

American Trucking Association

2200 Mill Road

Alexandria, VA 33314-5388

Phone: (703) 838-1763

Fax (703) 836-6070

Fully-formulated antifreeze contains balanced amounts of antifreeze, SCA, and buffering compounds, but does **not** contain 50 percent water. Fully-formulated coolant contains balanced amounts of antifreeze, SCA, and buffering compounds already premixed 50/50 with deionized water.

The following pages explain water, antifreeze, and SCA's and how to test antifreeze and SCA levels.

This section also contains information on cooling system maintenance and a coolant treatment chart that is used to determine the correct SCA service filter.

Alternative maintenance practices for cooling systems can be found in Cummins® Coolant Requirements and Maintenance, Bulletin 3666132.

Specifications

Automotive Applications

Coolant Capacity (engine only)

With EGR	33.1 liters [35 qt]
Without EGR	24 liters [25 qt]

Standard Modulating Thermostat

Temperature Range	82 to 93°C [180 to 200°F]

Maximum Coolant Pressure (exclusive of pressure cap - closed thermostat at the maximum no-load governed speed)

ISX CM871	434 kPa [63psi]
ISX CM870	400 kPa [38 psi]
ISX Without EGR and QSX	227 kPa [33 psi]

Coolant Alarm Activation Temperature

With EGR - Ratings Below 565 Horsepower	107°C [225°F]
With EGR - Ratings of 565/600 Horsepower	110°C [230°F]
Without EGR	107°C [225°F]

Maximum Allowable Top Tank Temperature

Ratings Below 565 Horsepower with EGR and without EGR Engines	107°C [225°F]
Ratings of 565/600 Horsepower with EGR only	110°C [230°F]

Minimum Recommended Top Tank Temperature

Minimum Temperature	70°C [160°F]

Minimum Allowable Draw Down

With EGR	11 Percent
Without EGR	2.4 liters [2.5 qt] or 10 Percent of System Capacity (whichever is greater)

Minimum Recommended Pressure Cap

ISX With EGR	103 kPa [15 psi]
ISX Without EGR	50 kPa [7 psi]

Minimum Fill Rate

Without Low-Level Alarm	19 liters/min [5 gpm]

Maximum Deaeration Time

Maximum Time	25 minutes

Fan-on Coolant Temperature

With EGR	99°C [210°F]
Without EGR	99°C [210°F]

Fan-on Intake Air Temperature

With EGR	93°C [200°F]
Without EGR	88°C [190°F]

Shutter Opening Temperature

Coolant - with EGR	96°C [205°F]
Coolant - without EGR	85°C [185°F]
Intake Air - with EGR	104°C [220°F]
Intake Air - without EGR	66°C [150°F]

Winterfronts

Air passage area	774 cm^2 [120 in^2]

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